

4. EMBEDDED TORIC RESOLUTION

The rational double point singularities are given by one of the following equations:

$$\begin{aligned} A_n : f(x, y, z) &= xy - z^{n+1}, \quad n \in \mathbb{N} \\ D_n : f(x, y, z) &= z^2 - x(y^2 + x^{n-2}), \quad n \in \mathbb{N}, n \geq 4 \\ E_6 : f(x, y, z) &= z^2 + y^3 + x^4 \\ E_7 : f(x, y, z) &= x^2 + y^3 + yz^3 \\ E_8 : f(x, y, z) &= z^2 + y^3 + x^5 \end{aligned}$$

Remark 4.1. Each member of ADE -singularities is Newton non-degenerate.

Embedded toric resolutions of ADE -singularities were studied in [18, 19]. Here differently we construct regular refinements of Gröbner fans of each members of them. This leads us their embedded toric resolutions [1, 2]. Additionally, these resolutions will be minimal which means that:

Definition 4.2. An embedded toric resolution is called minimal if

- i) The vectors in the regular refinement are irreducible.
- ii) The resolution graph obtaining from the refinement does not have any -1 curve means the self-intersection of each vertex is different from -1 (this gives us the resolution graph is minimal).

Note that the self-intersection of a vertex is computed as: The number $-s \in \mathbb{Z}_{\leq 0}$ with $s.v = \sum_{i=1}^m v_i$ is called the self-intersection of the vertex v in the graph where each v_i is adjacent to v in the regular refinement.

4.1. A_n -singularities. Consider the hypersurface $X = V(f) \subset \mathbb{C}^3$ where

$$f(x, y, z) = xy - z^{n+1}, \quad n \in \mathbb{N}$$

Given $v_1 \in (\mathbb{R}_{\geq 0})^3$, let $In_{v_1}(f) = f$. The polyhedral cone associated with v_1 is

$$C_{v_1}(f) = \langle (n+1, 0, 1), (0, n+1, 1) \rangle$$

Given $v_2 \in (\mathbb{R}_{\geq 0})^3$, let $In_{v_2}(f) = xy$. The polyhedral cone associated with v_2 is

$$C_{v_2}(f) = \langle (0, 0, 1), (n+1, 0, 1), (0, n+1, 1) \rangle$$

Given $v_3 \in (\mathbb{R}_{\geq 0})^3$, let $In_{v_3}(f) = -z^{n+1}$. The polyhedral cone associated with v_3 is

$$C_{v_3}(f) = \langle (1, 0, 0), (0, 1, 0), (n+1, 0, 1), (0, n+1, 1) \rangle$$

The Gröbner fan $G(X)$ of X and its Gröbner cones can be seen as

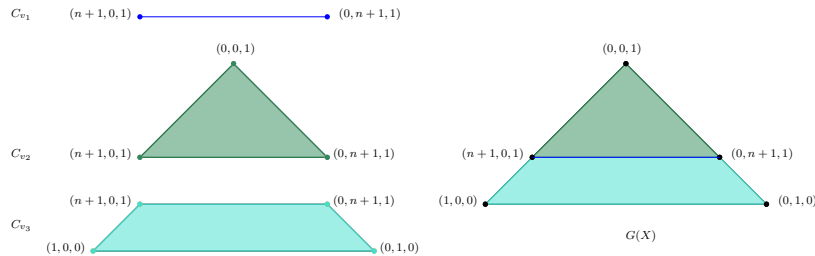


FIGURE 1. The Gröbner fan and its Gröbner cones for A_n -singularities